PROJECT SYNOPSIS

Resource-Efficient, Citation-Grounded Multilingual Document Intelligence for Private Enterprise Archives

A Retrieval-Augmented Generation system that answers questions over private, multilingual English & Indian-language document archives — with page-level citations and evidence-backed trust.

ENGLISH | HINDI | MARATHI | RAG | OCR | PGVECTOR

DOMAIN
Multilingual NLP & Information Retrieval

SYSTEM TYPE
Retrieval-Augmented Generation (RAG)

CORE STUDY
Multilingual vs. Translation vs. Hybrid Retrieval

OVERVIEW

Abstract

Organizations increasingly hold private documents in English and regional Indian languages, spread across native PDFs, DOCX files, and scanned images. Plain keyword search misses semantically related passages — a query for *"contract penalty"* can miss a clause phrased as *"liquidated damages"* — while general-purpose language models can produce fluent but unsupported answers, a serious risk in legal, medical, academic, and compliance settings.

This project designs and evaluates a **multilingual Retrieval-Augmented Generation (RAG) system** that retrieves evidence from private document libraries and produces answers with **page-level citations**. Its central research contribution is a controlled comparison of three retrieval strategies — direct multilingual embeddings, translation-to-English embeddings, and an adaptive hybrid dense-plus-keyword route — so that the system can select or combine strategies based on language, query, and document characteristics rather than assuming one approach fits every case.

The pipeline combines OCR-aware ingestion, semantic and lexical search, cross-encoder reranking, a lightweight answer generator, and an explicit **abstention mechanism** for when evidence is insufficient. The intended contribution is a reproducible evaluation of retrieval accuracy, citation faithfulness, language fidelity, OCR robustness, latency, and resource cost for English and selected Indian languages.

How it works, in plain terms

RAG stands for **Retrieval-Augmented Generation** and works in two stages: (1) **Retrieval** — find the most relevant passages from the uploaded documents, and (2) **Generation** — use a language model to write an answer using only that retrieved evidence, citing exactly where each claim came from.

1. User uploads a document (PDF, DOCX, or scanned image)
2. Text extraction or OCR, language detection, and page-aware chunking
3. Chunks embedded and stored in a searchable vector database (with metadata)
4. User asks a question in English or an Indian language
5. System retrieves and reranks the best evidence passages
6. Answer generated with document-name and page-number citations, or an abstention

THE GAP

Problem Statement

Existing multilingual RAG research usually concentrates on a single dimension at a time — translation quality, one low-resource language, multimodal retrieval, or a narrow legal corpus. There remains **insufficient evidence on how to build a trustworthy, resource-efficient system for heterogeneous private documents** in English and Indian languages, especially when source citations, OCR noise, language consistency, and constrained academic hardware all matter simultaneously.

Why this matters

- **Keyword search fails on paraphrase.** A query for “contract penalty” may miss a clause using “liquidated damages” — the words differ even though the meaning matches.
- **Unsupported LLM answers are risky.** A general-purpose model can answer confidently even with no source evidence — dangerous in legal, medical, academic, or compliance documents.
- **Multilingual retrieval is genuinely hard.** A Hindi question may need to retrieve Hindi text, English text, or both, and the best retrieval strategy differs by language, document type, OCR quality, and query type.

This is precisely why the comparison of retrieval strategies — not simply the construction of a chatbot — is the project's real research contribution.

PURPOSE

Aim

To design and evaluate a **citation-grounded multilingual RAG system** for private document collections that selects or combines retrieval strategies according to language, query, and document characteristics — producing answers that are accurate, verifiable, and affordable to run.

OBJECTIVES

What the project sets out to do

1. **Multi-format ingestion**
   Ingest native PDFs, DOCX files, and scanned documents through parsing, OCR, language identification, and page-aware chunking that preserves document name, page number, language, and paragraph position.

2. **Retrieval strategy comparison**
   Compare direct multilingual embedding, translation-based embedding, and hybrid dense-plus-BM25 retrieval to identify which approach best supports Indian-language and cross-lingual queries.

3. **Precision via reranking**
   Improve ranking quality using a multilingual cross-encoder reranker applied to the candidate passages returned by the retrieval stage.

4. **Cited, trustworthy generation**
   Generate concise answers with page-level document citations, and return an explicit insufficient-evidence response when the retrieved context does not support an answer.

5. **Rigorous, multi-dimensional evaluation**
   Evaluate retrieval quality, faithfulness, answer-language fidelity, latency, and resource use for English and selected Indian languages.

SYSTEM DESIGN

Proposed Methodology

STAGE 1 · INGESTION

Parsing, OCR & Chunking

Extract text and metadata from PDF and DOCX files; use **Tesseract OCR** for scanned documents. Detect language automatically, then split text into overlapping, page-preserving chunks so every chunk retains its source page number.

STAGE 2 · INDEXING

Vector + Lexical Store

Store chunk text, language, document ID, page number, and embeddings in **PostgreSQL with pgvector (HNSW)**. Maintain a parallel **BM25** lexical index alongside dense retrieval so exact-term matches are never lost to semantic search alone.

STAGE 3 · RETRIEVAL EXPERIMENT

Three Retrieval Conditions

(A) Direct multilingual — multilingual-E5 embeddings retrieve across languages natively.

(B) Translation-based — query or document is translated to English before embedding.

(C) Hybrid — dense-plus-BM25 retrieval combined with cross-encoder reranking.

An **adaptive router** may select the retrieval route based on language-confidence and query features.

STAGE 4 · GENERATION & TRUST

Grounded Answers with Abstention

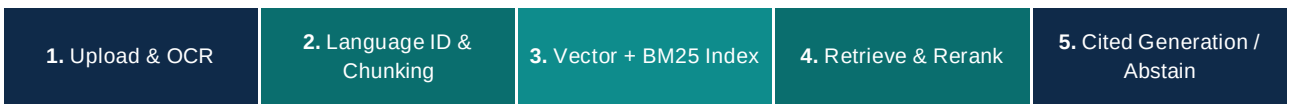
The generator receives only reranked evidence chunks and returns cited claims with document/page references. If retrieval confidence or evidence coverage is low, the system returns an **abstention** with a suggested query refinement instead of guessing.

STAGE 5 · DELIVERY

Application Architecture

A **FastAPI** backend handles asynchronous ingestion; a **Next.js** interface supports upload, querying, citation viewing, and streamed answers. Workspace isolation is an engineering feature of the delivery layer, not the central research claim.

End-to-end pipeline at a glance



MEASURING SUCCESS

Evaluation Plan

Dataset

A consented/private or public evaluation collection containing English and selected Indian-language documents — both native PDFs and OCR scans — with manually checked question–answer–evidence triples.

Retrieval Metrics	
Recall@k	Whether the correct evidence passage appears in the top-k retrieved results.
MRR@10	How highly the first correct passage is ranked, averaged across queries.
nDCG@10	Ranking quality that rewards placing the best evidence closer to the top.
Per-language results	Whether retrieval quality holds steady across English, Hindi, Marathi, and other languages.

Answer Metrics	
Human correctness	Whether the generated answer is factually correct, judged by a human rater.
Citation precision/recall	Whether cited pages truly support the claim, and whether supporting pages are missed.
Faithfulness / groundedness	Whether every claim in the answer is actually backed by retrieved evidence.
Abstention precision	Whether the system correctly declines to answer unanswerable questions.

Operational & Robustness Metrics	
Latency & throughput	End-to-end response time and how many queries the system can serve.
GPU/CPU memory & cost	Resource footprint per query, relevant for constrained academic hardware.
Format / OCR robustness	Performance across PDF, DOCX, and scanned formats, and under OCR noise.
Cross-lingual pairs	Accuracy when the query language differs from the document language.

Baselines

- **Vanilla dense multilingual RAG** — direct multilingual embeddings only.
- **Translation-only RAG** — translate to English before retrieval, no hybrid route.
- **Hybrid retrieval without reranking or abstention** — isolates the contribution of reranking and evidence-sufficiency checks.

OUTCOMES

Expected Contributions

1. A controlled comparison of multilingual, translation-based, and hybrid retrieval for English and Indian-language private documents.
2. A page-citation and evidence-sufficiency mechanism for more trustworthy, verifiable answers.
3. An evaluation protocol covering retrieval, groundedness, language behaviour, OCR robustness, latency, and cost.
4. A deployable, open-source-oriented architecture suitable for constrained academic hardware.

SCOPE

Scope and Limitations

The first implementation should claim support only for languages actually evaluated — for example English, Hindi, Marathi, and one or two additional Indian languages. Table and chart understanding requires a dedicated vision-language component and should be presented as **future work** unless implemented and tested. The system assists document search and **does not provide legal, medical, or compliance advice.**

BIBLIOGRAPHY

Key References

- Park, J. and Lee, H. (2025). *Investigating Language Preference of Multilingual RAG Systems.* Findings of ACL 2025.
- Kabir, M. R. et al. (2025). *LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval.*
- Xu, J., Zhu, J., and Ba, Y. (2025). *M3-RAG: Unified Multimodal and Multilingual Retrieval-Augmented Generation.*
- Iscan, C., Ozara, M. F., and Akbulut, A. (2024). *Enhancing RAG Pipeline Performance with Translation-Based Embedding Strategies for Non-English Documents.*
- Zoughi, T. et al. (2026). *Hybrid Layered Retrieval and Task-Aware Embeddings for Efficient Persian RAG Systems.*

This synopsis consolidates and extends the project brief with an explicit problem statement, methodology breakdown, evaluation protocol, and scope boundaries, aligned to the seven surveyed papers referenced above.